MOBILE APPLICATION DEVELOPMENT

IMPORTANT QUESTIONS — ALL 5 UNITS

Polytechnic 6th Semester | Based on supplied 2023–2025 papers and your syllabus | ★ = high priority

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UNIT I — INTRODUCTION TO ANDROID

★ What is Android? Explain Android Platform Architecture with a suitable diagram.

★ Explain Dalvik Virtual Machine (DVM) and its role in Android.

★ Explain the Android SDK and the steps to install and configure the Android development environment.

★ Explain the steps involved in creating your first Android application.

• Explain the anatomy and basic project structure of an Android application.

★ Explain the Android Manifest XML file and its purpose.

• Define Android API levels. Explain Android versions and their version names.

• Write a short note on DDMS (Dalvik Debug Monitor Server).

UNIT II — ANDROID APPLICATION DESIGN ESSENTIALS

★ Explain the basic building blocks or components of an Android application.

★ What is an Intent? Explain the types of Intents with suitable examples.

★ Explain the Activity Life Cycle with a diagram.

★ What is a Service in Android? Explain its types and life cycle.

★ Explain Broadcast Receivers and the different ways to register them.

★ Explain Content Providers and how they work.

• Explain Application Context and its uses.

★ Explain Android Manifest settings, Intent Filters and permissions.

• Explain the steps involved in creating an Android application.

UNIT III — ANDROID USER INTERFACE DESIGN ESSENTIALS

★ Explain Android UI components and controls with examples.

★ Explain different types of Android layouts: Linear, Relative, Frame and Grid Layout.

★ Explain Android Menus and their types: Options Menu, Context Menu and Popup Menu.

• Explain User Interface Design principles in Android.

★ Explain GridView and CardView with examples.

• Explain Toast messages and ViewHolder.

- Explain event handling in Android.
- Explain drawable resources and their uses.
- Explain Fragments and their role in Android UI.
- Explain drawing and animation in Android.

UNIT IV — TESTING, PUBLISHING AND APPLICATION RESOURCES

★ Explain Android application testing and the different testing methods.

★ Explain the steps involved in publishing an Android application.

★ What are Shared Preferences? Explain their uses with an example.

★ Explain Android resources and resource management in a hierarchy.

- Explain drawable, string, color, layout and other Android resources.
- Explain Android project resources and the R class.
- Explain the Android application build and deployment process.

UNIT V — COMMON ANDROID APIs

★ Explain SQLite Database in Android and its importance.

★ Explain CRUD operations (Create, Read, Update and Delete) in SQLite with examples.

★ What is a Content Provider? Explain how data is shared between applications.

★ Explain Threads, Worker Threads, Handler and Runnable in Android.

★ Explain Notifications and Services in Android.

- Explain Android storage options and data storage APIs.
- Explain how to consume JSON data from a web service.
- Explain Android Networking APIs and Web APIs.
- Explain Android Telephony APIs and their applications.
- Explain Broadcast Receivers and their use in receiving system events.

Study note: *The starred questions are priority topics based on recurrence in the provided papers and relevance to your syllabus. They are not a guarantee of the questions that will appear in the examination. Prepare diagrams, examples and steps where appropriate.*

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